



scDEBART: Predicting *in silico* Single-Cell Perturbation Responses via Large-Scale Differential Expression Learning

Jieun Sung, Wankyu Kim*

Systems Pharmacology Lab, Department of Life Science, Ewha Womans University, Seoul, Republic of Korea

Presenter contact: jieunsung@ewha.ac.kr · *Correspondence: wkim@ewha.ac.kr



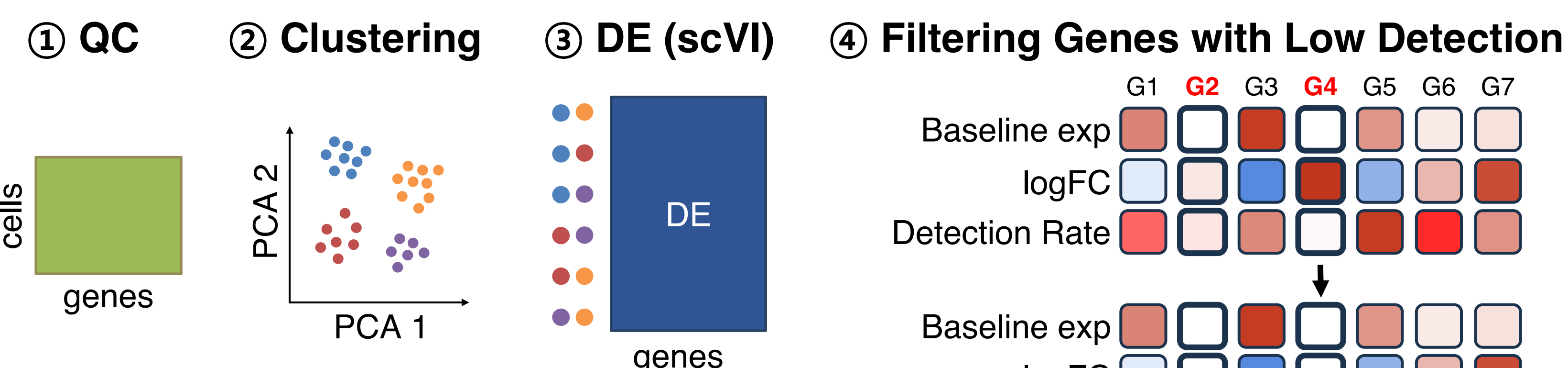
EWHA WOMANS
UNIVERSITY

Research Problem

- Single-cell foundation models (scGPT, GeneFormer) do not outperform simple linear baselines on perturbation prediction (Ahlmann-Eltze et al., 2025)
- Limitation 1) targets on **dropout-prone absolute expression** → unstable post-hoc fold changes 2) pretraining reconstructs **static within-cell co-expression**, not how genes co-regulate under change
- scDEBART**: pretrain to predict **population-level logFC conditioned on basal expression**, not absolute expression → learns **regulatory co-variation across millions of contexts** as an inductive bias for perturbation

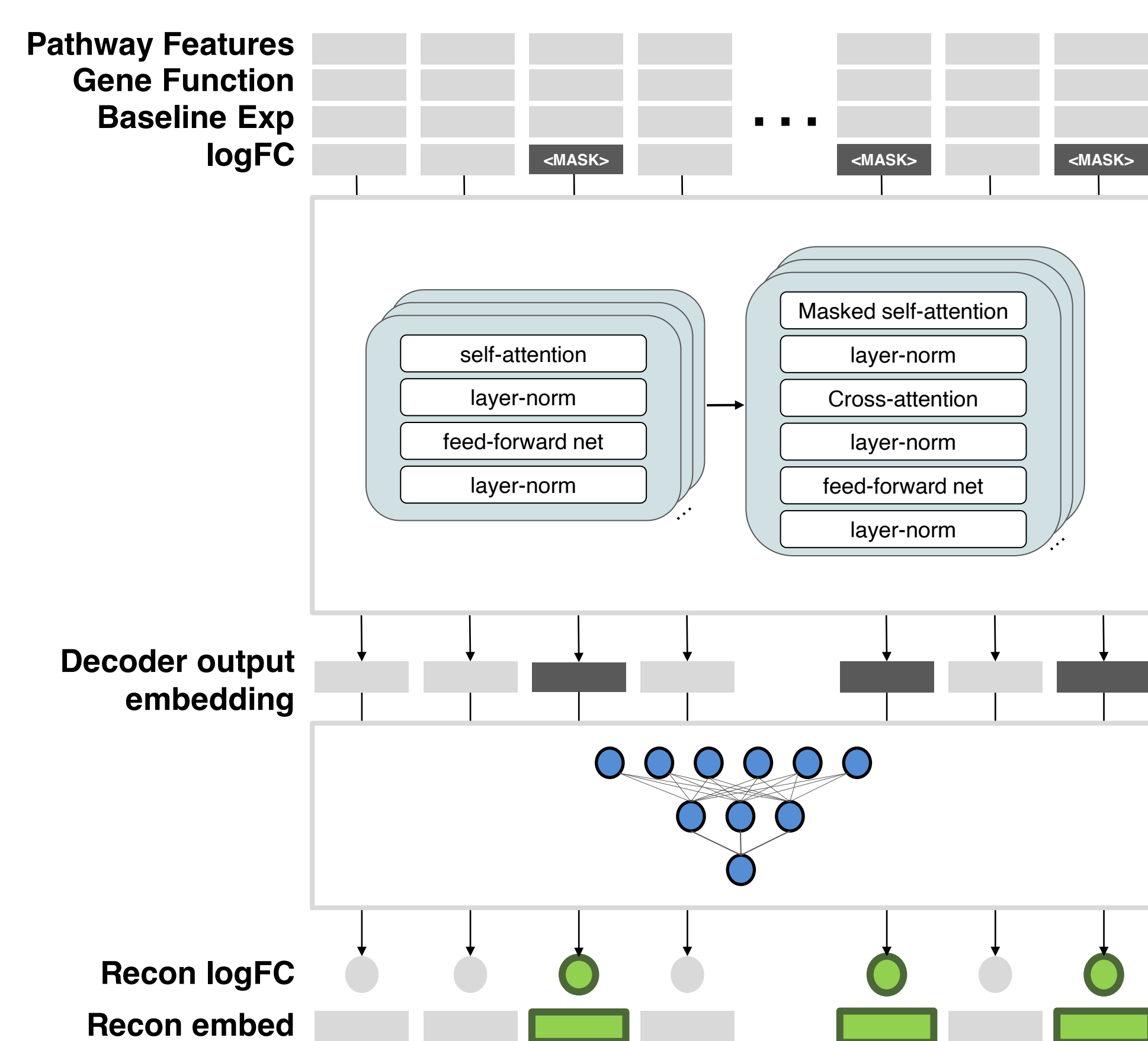
Data & Methods

Constructing a Large-Scale DE Corpus from CELLxGENE



- Filtered CELLxGENE to 617 datasets comprising 66.6M cells across 61 tissues
- Grouped cells within each dataset using SECUER density-based clustering, yielding 98,753 clusters with a median of 278 cells per cluster
- Computed cluster-pair differential expression (DE) using scVI posterior sampling, retaining genes with DE probability ≥ 0.8 , detection rate ≥ 0.3 , and IFCI ≥ 1.5
- 6.28M DE profiles derived from 66.6M cells, covering 17,944 genes**

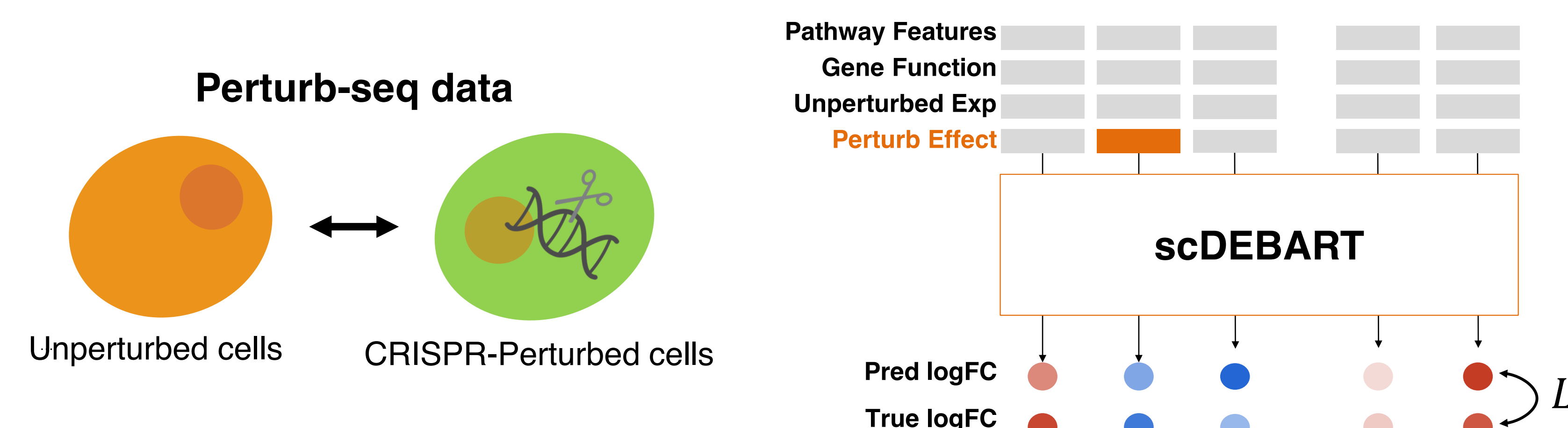
scDEBART: Architecture & Training



- Input (per gene token):** Fuses 4 features
 - MSigDB C2 pathway membership
 - BioBERT embedding (gene summary)
 - scVI logFC
 - log-baseline expression
- Model:** BART — 8 enc + 4 dec, 12 heads, d=768
- Pretrain** (large DE corpus)
Mask 15% of gene logFCs → reconstruct logFC + embedding
- Fine-tune** (Perturb-seq) : ± 10 perturbation indicator (activate / inhibit), Loss: weighted-MSE + correlation

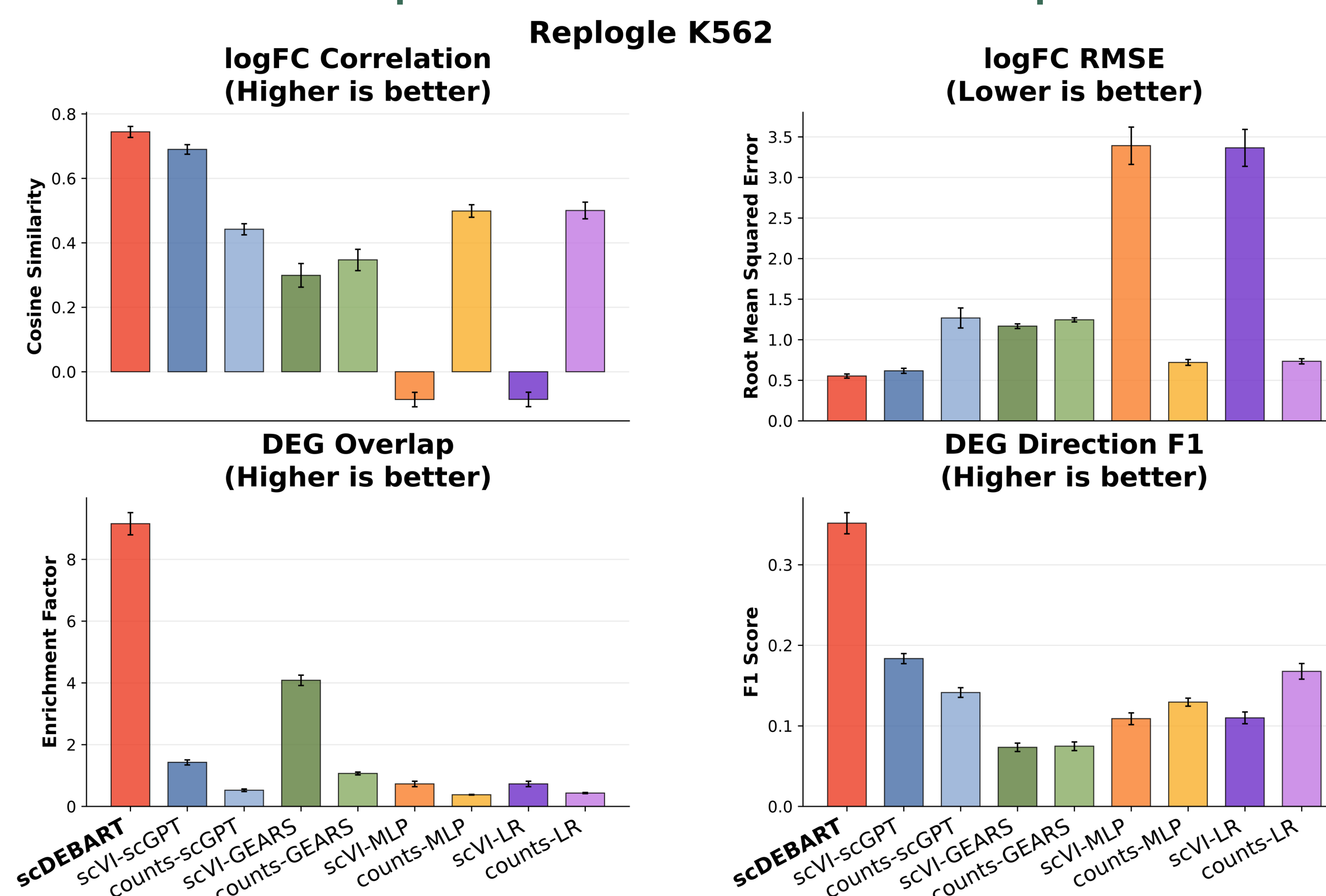
1. Genetic Perturbation Response Prediction

Extracting Reliable Perturbation Signatures from Perturb-seq



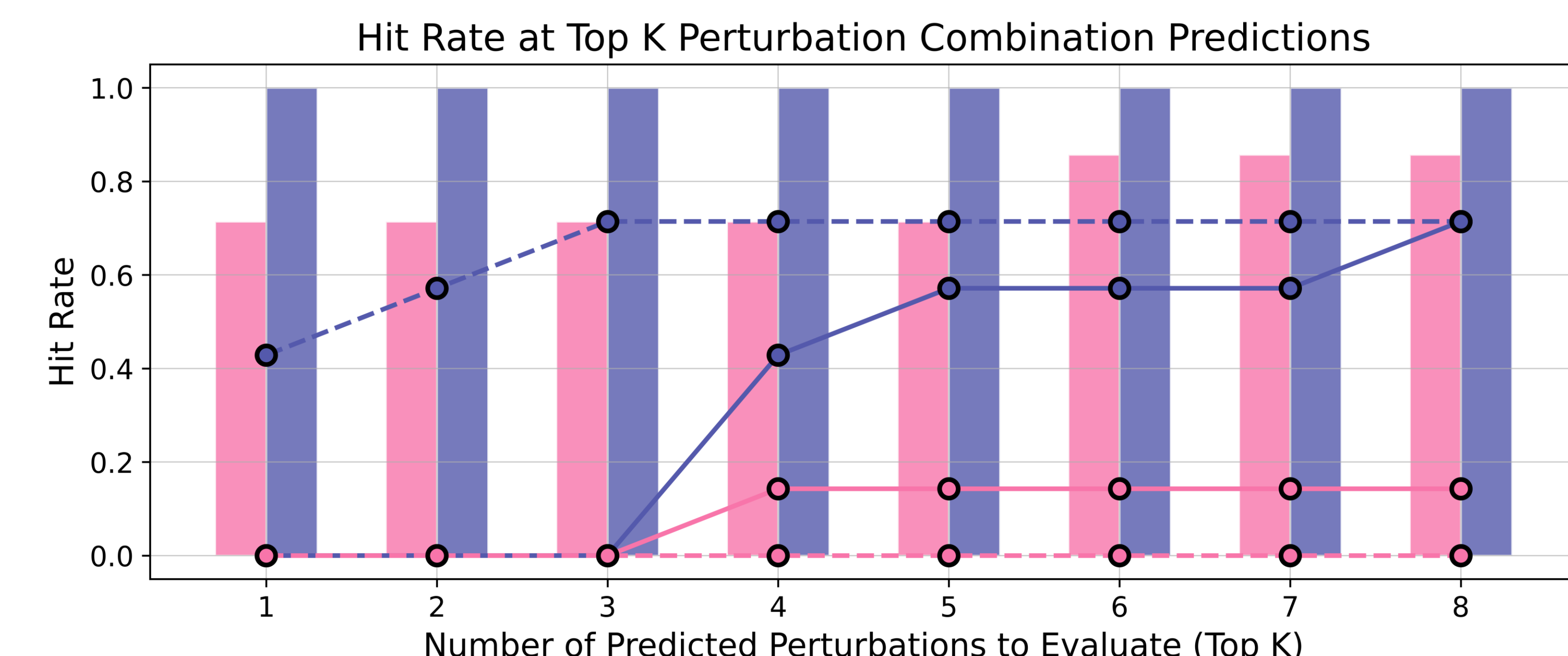
- QC & target expression:** retain cells with effective target silencing or activation
- Perturbation score (PS):** retain cells above a dataset-specific cutoff (Song et al., 2025) to remove weakly perturbed or unperturbed cells
- Cluster coherence:** retain cells closer to their perturbation centroid than to controls, yielding tighter perturbation-specific clusters
- DE computation:** derive per-gene logFC vs. control via scVI posterior sampling

scDEBART Outperforms Baselines Across Multiple Metrics



- Comparative analysis across **5 Perturb-seq datasets** (Norman K562, Replogle K562/RPE1, Nadig HepG2/Jurkat)
- Replogle K562 shown as representative, evaluated on 4 complementary metrics:**
 - logFC correlation:** do predicted genes move in the same direction as the truth?
 - logFC RMSE:** how close are the predicted change sizes?
 - DEG Overlap:** are the real top DEGs ranked at the top?
 - DEG direction:** are up/down calls correct (1.2-fold)?

2. Reverse Perturbation Identification

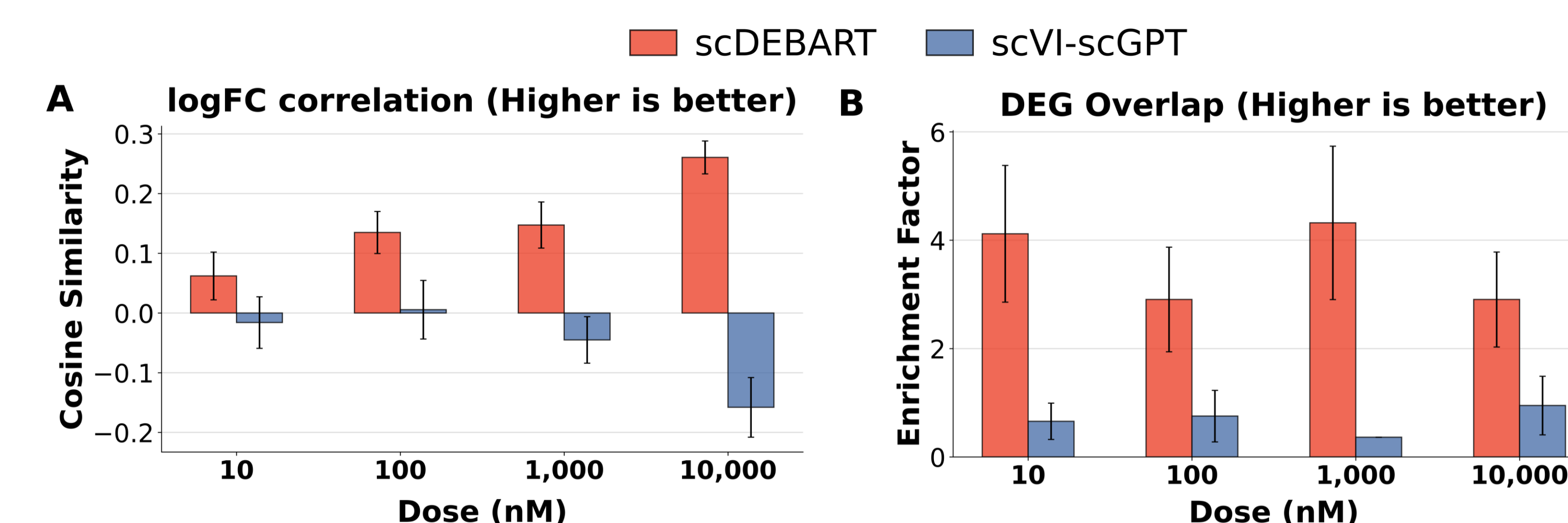


Correct perturbation combination predicted
At least one gene in the perturbation combination predicted

scDEBART scGPT GEARS

- Task:** given an observed logFC profile, infer which gene(s) were perturbed
- scDEBART **top-1 exact match 71.4%** vs **0%** for scGPT & GEARS; partial match **100%** (GEARS 42.9%, scGPT 0%)
- predicted logFC profiles are **perturbation-specific** enough to invert, where baselines fail

3. Cross-modal Transfer: Genetic → Drug Perturbation



- A Replogle-K562 genetic-perturbation model was applied to drug responses without drug-specific training by simulating inhibitor targets as multi-gene knockdowns.
- Dose-dependent performance:** Similarity increased from cosine 0.04 to 0.30 across 10–10,000 nM, with EF 2.91–4.32, consistently outperforming scVI-scGPT.

Key Findings

- Pretraining:** population-level logFC conditioned on basal expression → learns regulatory co-variation (6.28M DE profiles / 66.6M cells), with dropout pruning and low-detection filtering.
- Genetic perturbation:** best DEG recovery across 5 Perturb-seq datasets.
- Dropout artifact:** baselines rank low-detection genes as top DEGs; our filtering removes these false positives — a step skipped by prior work.
- Take-home:** dropout-robust, change-based supervision at scale is a stronger inductive bias for perturbation prediction — a step toward virtual-cell models.